Which of the five boroughs comprising the New York City are most similar?

Daniel Costelloe

April 11, 2019

1. Introduction

1.1 Background

One idea suggested by the instructor, is to compare two cities and see how similar or dissimilar they are. In the previous exercise, neighborhoods in Manhattan were examined. However, New York City has five boroughs, corresponding to the five counties within New York City. The five boroughs of New York City were each themselves their own city at one time. Five cities came together to form what is now New York City. The five boroughs are Bronx, Brooklyn, Manhattan, Queens, and Staten Island. Respectively these are each a county: Bronx County, Kings County, New York County, Queens County, and Richmond county. Many cities are within a county, rather than comprising of counties. The boroughs of New York City can be treated as cities for the purposes of the suggested question of examining how similar or dissimilar cities can be.

1.2 Problem

Each borough has its own characteristics and reputation. At the same time, New York City is a dynamic place. As the boroughs change, their characteristics and therefore their similarities and dissimilarities also change. Heuristics used even by New Yorkers to group the boroughs may be misconceived or out of date at any given time. Which of the boroughs of New York City, former cities in their own right, are most similar.

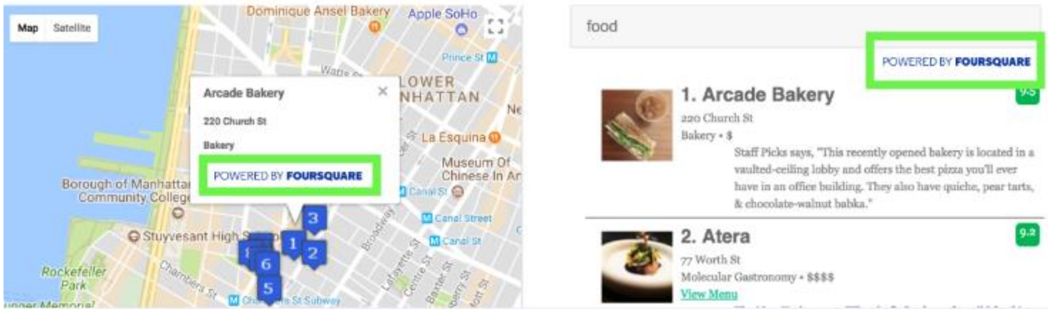
1.3 Interest

There are many possible applications of knowing which boroughs are most similar. One case might be New Yorkers happy with their current location, yet believing they will have to move soon. Knowing which boroughs are similar or dissimilar to the one they live in and like (or dislike) can help guide them in choosing which other locations to examine as a possible new residence. In addition, real estate brokers armed with data showing these similarities/dissimilarities might be more able to persuade clients to consider looking at areas they might not otherwise consider. Recently residents of two boroughs believed their neighborhood characteristics might change drastically, and many considered moving or in some cases moved. Neighborhoods of Brooklyn were faced with the L train subway shutting down for an extended period of time. Neighborhoods in Queens were faced with the prospect of Amazon establishing a head quarters there, and it was believed rents would climb drastically, and that the already crowded subways would become much more crowded. (At the last minute the L train had its repair plan changed to allow at least limited train service to continue, and Amazon backed out of establishing a headquarters in Queens.) Brooklyn and Queens are geographically contiguous and are often lumped together as “outer boroughs”, i.e. not Manhattan, but are they similar, or are other noncontiguous boroughs, even counter intuitively Manhattan, more similar?

2. Data

2.1 Data Sources

2.1.1 As required by the assignment, FourSquare data was used. Using API calls, FourSquare data was obtained providing the categories for venues found within 500 meters of latitude and longitude for each neighborhood. These data were used for ranking the most common types of venues within a neighborhood. By pulling data from neighborhoods within each borough, similar to a stratified sample, we ensure representation from each section of each borough. A limit of 30 was used for the venues form each neighborhood. FourSquare suggests using an image as a citation:



2.1.2 For each neighborhood, the neighborhood name and its corresponding latitude and longitude where obtained from a csv file which the instructor downloaded from https://geo.nyu.edu/catalog/nyu_2451_34572, which requests to be cited as http://hdl.handle.net/2451/34572. This is from the NYU Faculty Digitial Archive, FDACommunities & Collections, Division of Libraries, Spatial Data Repository, 2014 New York City Neighborhood Names. The website provides this description of the data:

1. Title: 2014 New York City Neighborhood Names
2. Issue Date: 12-Apr-2016
3. Description: This New York City Neighborhood Names point file was created as a guide to New York City's neighborhoods that appear on the web resource, "New York: A City of Neighborhoods." Best estimates of label centroids were established at a 1:1,000 scale, but are ideally viewed at a 1:50,000 scale.
4. Publisher: New York (City). Department of City Planning
5. Collection: Bytes of the Big Apple

2.1.3 The python geolocator.geocode was used to obtain latitude and longitude coorindates for New York City and the Boroughs in general. This functionality is documented at https://geopy.readthedocs.io/en/stable/

2.2 Data Cleaning

2.2.1 The data did not have unique names for the neighborhoods. Some neighborhood names existed in more than one borough. To create unique names, a comma and the borough name was appended to the neighborhood name. This resulted in the column Neighborhood having unique values.

2.2.2 In a previous exercise, using Toronto data, one venue category had the value of 'Neighborhood', which post dummy variable transformation, had the unfortunate effect of creating a second column named 'Neighborhood', resulting in two columns in the same data frame with the same column name. To guard against this eventuality, any Venue Category values of Neighborhood were changed to be Neighborhood Category to differentiate the resulting column.

3. Methodology

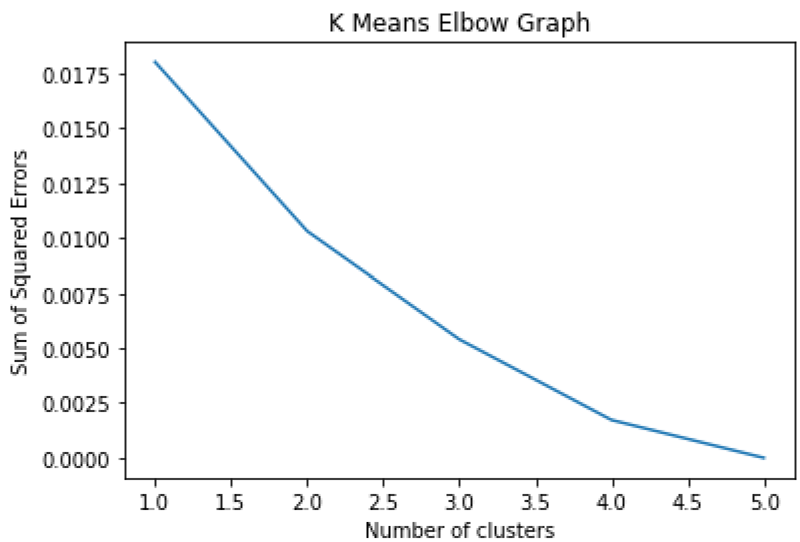
3.1 K Means clustering was applied to group the boroughs into clusters based on the most common venue type in each borough.

3.2 Numerical variables were created representing the relative frequency of each venue category within a borough.

3.3 Sorting this data produced a listing of the most common venue types in each borough.

Borough	Number_1_Venue_Type	Number_2_Venue_Type	Number_3_Venue_Type	Number_4_Venue_Type	Number_5_Venue_Type	Number_6_Venue_Type	Number_7_Venue_Type	Number_8_Venue_Type	Number_9_Venue_Type	Number_10_Venue_
Bronx	Pizza Place	Deli / Bodega	Supermarket	Pharmacy	Chinese Restaurant	Donut Shop	Bus Station	Sandwich Place	Italian Restaurant	Spanish Restau
Brooklyn	Cluster Labels	Pizza Place	Coffee Shop	Bar	Italian Restaurant	Grocery Store	Café	Deli / Bodega	Bakery	Ice Cream
lanhattan	Cluster Labels	Coffee Shop	Park	American Restaurant	Italian Restaurant	Café	Pizza Place	Bakery	Gym	Cockta
Queens	Pizza Place	Deli / Bodega	Chinese Restaurant	Bakery	Pharmacy	Donut Shop	Bank	Italian Restaurant	Sandwich Place	Coffee
Staten Island	Cluster Labels	Pizza Place	Bus Stop	Deli / Bodega	Italian Restaurant	Bagel Shop	Pharmacy	Bank	Sandwich Place	Chinese Restau

3.4 A K-Means "Elbow" graph was produced graphing the number of clusters by the sum of the squared errors for each clustering.



3.5 Although the elbow graph would suggest that four clusters have meaningful differentiation, the purpose of the investigation is to see which boroughs are similar and dissimilar, this purpose would be defeated by putting almost each borough in its own cluster, or by putting most boroughs in one cluster. A Goldilocks, just right, number of three clusters was chosen for this reason.

3.6 The clusters were assigned to the boroughs based on most frequent venue category.

Borough	Number_1_Venue_Type	Number_2_Venue_Type	Number_3_Venue_Type	Number_4_Venue_Type	Number_5_Venue_Type	Number_6_Venue_Type	Number_7_Venue_Type	Number_8_Venue_Type	Number_9_Venue_Type
Bronx	Pizza Place	Deli / Bodega	Supermarket	Pharmacy	Chinese Restaurant	Donut Shop	Bus Station	Sandwich Place	Italian Restaurant
Brooklyn	Pizza Place	Coffee Shop	Bar	Italian Restaurant	Grocery Store	Café	Deli / Bodega	Bakery	Ice Cream Shop
Manhattan	Coffee Shop	Park	American Restaurant	Italian Restaurant	Café	Bakery	Pizza Place	Cocktail Bar	Gym
Queens	Pizza Place	Deli / Bodega	Chinese Restaurant	Bakery	Pharmacy	Donut Shop	Bank	Italian Restaurant	Sandwich Place
Staten Island	Pizza Place	Bus Stop	Deli / Bodega	Italian Restaurant	Bagel Shop	Pharmacy	Sandwich Place	Bank	Grocery Store

3.7 The latitude and longitude for each borough was merged into the data to enable representing the clusters on a map.

3.8 Labels were assigned to the clusters representing both the boroughs included in the cluster, and the most common venue types distinctive to each cluster.

4. Results

4.1 The following boroughs were grouped into these clusters:

4.1.1 The Bronx and Queens

4.1.2 Brooklyn and Manhattan

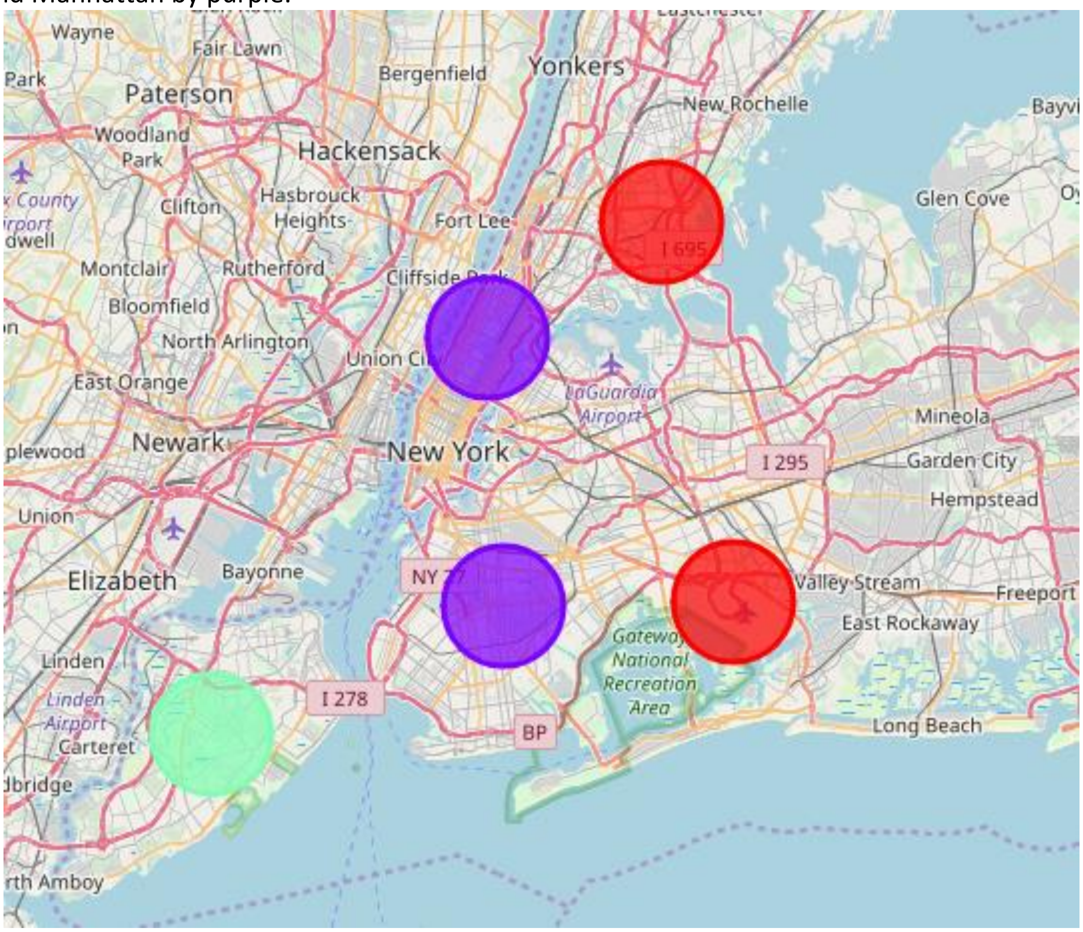
4.1.3 Staten Island

4.2 Every borough except Manhattan had Pizza Place as the number one most common venue type. Both the Bronx and Queens had Deli/Bodega as the second most common venue type. Staten

Island had Bus Stop as the second most common venue type. Brooklyn had Coffee Shop as the second most common venue category. Manhattan had Coffee Shop and Park as its most common venue type.

Cluster_Name	Borough	Number_1_Venue_Type	Number_2_Venue_Type	Number_3_Venue_Type	Number_4_Venue_Type	Number_5_Venue_Type	Number_6_Venue_Type	Number_7_Venue_Type	Number_8_Venue_Type	Number_9_Venue_Type
Bronx and Queens (Pizza Place and Deli / Bodega)	Bronx	Pizza Place	Deli / Bodega	Supermarket	Pharmacy	Chinese Restaurant	Donut Shop	Bus Station	Sandwich Place	Italian Restaurant
Bronx and Queens (Pizza Place and Deli / Bodega)	Queens	Pizza Place	Deli / Bodega	Chinese Restaurant	Bakery	Pharmacy	Donut Shop	Bank	Italian Restaurant	Sandwich Place
Brooklyn and Manhattan (Coffee Shop)	Brooklyn	Pizza Place	Coffee Shop	Bar	Italian Restaurant	Grocery Store	Café	Deli / Bodega	Bakery	Ice Cream Shop
Brooklyn and Manhattan (Coffee Shop)	Manhattan	Coffee Shop	Park	American Restaurant	Italian Restaurant	Café	Bakery	Pizza Place	Cocktail Bar	Gym
Staten Island (Pizza Place [illegible]	Staten Island	Pizza Place	Bus Stop	Deli / Bodega	Italian Restaurant	Bagel Shop	Pharmacy	Sandwich Place	Bank	Grocery Store

4.3 A choropleth map was created showing the clusters. Each cluster is represented by a different color. Staten Island is represented by light green, the Bronx and Queens by red, and Brooklyn and Manhattan by purple.



5. Discussion

5.1 The Bronx and Queens were the most similar boroughs. Not only did they match on their first two categories, but within the next four categories (Pizza and Deli), they both had: Chinese Restaurant, Pharmacy, and Donut Shop. Staten Island stood out for having Bus Stop in its top two categories and Bagel Shop in its top five. Brooklyn and Manhattan both had Coffee Shop in their top two, Italian Restaurant as fourth, and Café ranked as either fifth or sixth.

5.2 I found it surprising that Brooklyn was found to be similar to Manhattan. For many years, Manhattan was considered what people thought of as New York, and the other boroughs were lumped together as “outer boroughs” or “bridge and tunnel” (based on methods of travel to Manhattan). The gentrification of many Brooklyn neighborhoods in recent years has changed the character of Brooklyn, as any long-time resident of the city can attest to. I was also surprised by how similar the Bronx and Queens are. The are both Pizza, Deli, Chinese Restaurant, and Donut Shop. These sound for the most part like New York’s versions of fast food. Staten Island has often been thought of as distinct among the non-Manhattan boroughs. That Bus Stop ranked in the top two may simply be an indication of the lack of an extensive subway system in Staten Island. However, even if Bus Stop were removed, Staten Island would still be the only borough with Bagel Shop and Grocery Store in its top eight most common venues (each moving up in frequency if Bus Stop were removed).

5.3 New York City is ethnically diverse. It might be worth exploring if some ethnic restaurants were grouped into larger categories, e.g. South Asian instead of Indian and Thai, if the diversity would become more apparent in the category rankings. Coffee and Café might also be grouped into one category, this would allow more categories to reach the top ten categories.

6. Conclusions

6.1 People who would like to live in Manhattan, but are deterred by the high cost of living in Manhattan might consider Brooklyn. As a Queens resident, I was surprised how very similar Queens was compared to the Bronx. People who like and know one of these boroughs should become more familiar with the other, and consider it if looking to relocate. People who do not like the other boroughs should explore Staten Island for its distinctive quality.